Netflix Movies Data Analysis Report

Project Title: Netflix Movies Data Analysis using Python

1. **Introduction**

Netflix is one of the world's leading entertainment streaming platforms. It was founded on August 29, 1997, in Scotts Valley, California, by Reed Hastings and Marc Randolph. The company initially operated as a DVD-by-mail rental service, allowing customers to rent movies online and receive them through postal delivery. The idea was inspired by Hastings' experience of paying a late fee at a traditional video rental store, while Randolph played a key role in developing Netflix's early business model, branding, and user experience.

In 2007, Netflix transformed the entertainment industry by launching its online streaming service, enabling subscribers to watch movies and TV shows instantly over the internet. This shift from DVD rentals to digital streaming helped the company meet changing consumer preferences and take advantage of increasing internet accessibility.

Netflix expanded internationally in 2010, beginning with Canada, and by 2016, its streaming service was available in more than 190 countries. Today, the platform offers a vast collection of movies, TV series, documentaries, and games across multiple genres and languages, serving audiences worldwide.

As of 2024, Netflix reported approximately $10 billion in revenue during the third quarter and a net profit of around $2.4 billion. The company has over 283 million paid subscribers globally, making it one of the largest and most successful streaming platforms in the world.

This project analyzes the Netflix Movies and TV Shows dataset using Python and Exploratory Data Analysis (EDA) techniques. The objective is to understand content distribution, identify trends in genres, ratings, release years, countries, and content types, and generate valuable insights through data visualization.

2. **Problem Statement**
 Netflix is widely recognized for its use of **Data Science, Artificial Intelligence (AI), and Machine Learning (ML)** to deliver personalized content recommendations and improve the user experience. By analyzing user behavior, viewing history, and content preferences, Netflix helps subscribers discover movies and TV shows that match their interests. In this project, we analyze a dataset containing **more than 9,000 Netflix movies and TV shows**. As a data analyst, the objective is to explore the dataset, identify meaningful patterns, and generate insights that support data-driven business decisions.
    1. What is the most frequent genre of movies released on Netflix?
    2. What genres has highest votes?
    3. What movie got the highest popularity? What's its genre
    4. What movie got the lowest popularity? What's its genre?
    5. Which year has the most filmed movies?

3. **Objectives**
    - Perform data cleaning and preprocessing.
    - Explore the Netflix dataset using Exploratory Data Analysis (EDA).
    - Visualize important trends.
    - Identify the most common genres, highest votes, popularity, lowest popularity, year.
    - Understand content growth over time.

4. **Dataset Description**
 The dataset contains information about Netflix Movies.
 Features
    - Release_Date
    - Title
    - Overview
    - Popularity
    - Vote_count
    - Vote_Average
    - Original_Language

- Genre
- Poster_Url

5. Tools and Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Excel/Kaggle

6. Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records.
- Checked missing values.
- Filled or removed null values.
- Converted data into appropriate data types.
- Cleaned inconsistent records.
- Prepared data for visualization.

7. Exploratory Data Analysis

The following analyses were performed:

- Movies analysis
- Release year analysis
- Rating distribution
- Genre analysis
- Year-wise content growth

8. Visualizations

The project includes:

- Count Plot
- Bar Chart
- Histogram
- Pie Chart
- Heatmap
- Box Plot

These visualizations help identify trends and compare different categories.

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9. Key Findings

- Movies are more numerous than TV Shows.
- The United States contributes the largest amount of Netflix content.
- TV-MA is one of the most common content ratings.
- Drama and International Movies are among the most popular genres.
- Netflix rapidly increased its content library after 2015.
- Most movies have durations between 80 and 120 minutes.

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10. Conclusion

The analysis shows that Netflix focuses heavily on movies while continuously expanding its global content library. Data visualization makes it easier to understand audience preferences, production trends, and content distribution across countries and genres.

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11. Future Improvements

- Build a recommendation system.

- Perform sentiment analysis using descriptions.
- Create an interactive dashboard with Power BI or Tableau.
- Deploy the project using Streamlit.

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12. Learning Outcomes

Through this project, I learned:

- Data Cleaning
- Data Visualization
- Exploratory Data Analysis (EDA)
- Python Programming
- Working with real-world datasets
- Finding business insights from data